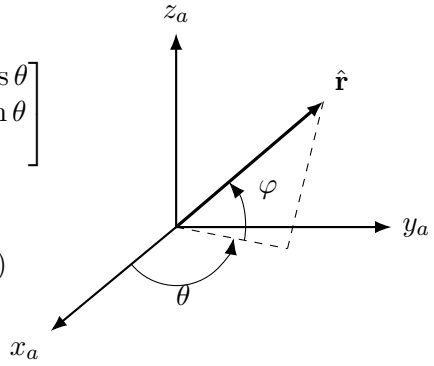
Quantitative measure of the force model

Step 1

1.1 Unit direction $\hat{\mathbf{r}}(\varphi, \theta)$

An arbitrary unit direction $\hat{\mathbf{r}}$ is described in the actuator-frame basis $\{\hat{\mathbf{x}}_a,\ \hat{\mathbf{y}}_a,\ \hat{\mathbf{z}}_a\}$:

$$\hat{\mathbf{r}}(\varphi, \theta) = \cos\varphi\cos\theta\,\hat{\mathbf{x}}_a + \cos\varphi\sin\theta\,\hat{\mathbf{y}}_a + \sin\varphi\,\hat{\mathbf{z}}_a = \begin{bmatrix} \cos\varphi\cos\theta \\ \cos\varphi\sin\theta \\ \sin\varphi \end{bmatrix}$$



$$\begin{cases} \varphi \in [-90^\circ,\ 90^\circ] & \text{elevation (from the } x_a\text{–}y_a \text{ plane)} \\ \theta \in [0^\circ,\ 360^\circ) & \text{azimuth (from the } +x_a \text{ axis, about } z_a) \end{cases}$$

$$\|\hat{\mathbf{r}}\| = 1$$

1.2 Current I

The force vector produced at position $\mathbf{p}_i$ by the current $\mathbf{I}$:

$$\mathbf{f}_i = \begin{bmatrix} f_{i1} \\ f_{i2} \\ f_{i3} \end{bmatrix}, \qquad f_{ik} = \mathbf{I}^\mathsf{T}\,{}^F\hat{\mathbf{H}}_I^\mathsf{T}\,\mathbf{L}_k\Big(\mathbf{p}_i/\hat{\ell},\,\hat{\mathbf{e}}\Big)\,{}^F\hat{\mathbf{H}}_I\,\mathbf{I}, \qquad k = 1, 2, 3$$

Let

$$\mathbf{I} = \sqrt{\|\mathbf{f}_i\|}\,\hat{\mathbf{I}}, \qquad \mathbf{f}(\mathbf{p}_i, \mathbf{I}) = \|\mathbf{f}_i\|\,\hat{\mathbf{r}}$$

Substituting into the force model:

$$\mathbf{f}_i = \begin{bmatrix} \Big(\sqrt{\|\mathbf{f}_i\|}\,\hat{\mathbf{I}}\Big)^\mathsf{T}\,{}^F\hat{\mathbf{H}}_I^\mathsf{T}\,\mathbf{L}_1\Big(\mathbf{p}_i/\hat{\ell},\hat{\mathbf{e}}\Big)\,{}^F\hat{\mathbf{H}}_I\,\Big(\sqrt{\|\mathbf{f}_i\|}\,\hat{\mathbf{I}}\Big) \\ \Big(\sqrt{\|\mathbf{f}_i\|}\,\hat{\mathbf{I}}\Big)^\mathsf{T}\,{}^F\hat{\mathbf{H}}_I^\mathsf{T}\,\mathbf{L}_2\Big(\mathbf{p}_i/\hat{\ell},\hat{\mathbf{e}}\Big)\,{}^F\hat{\mathbf{H}}_I\,\Big(\sqrt{\|\mathbf{f}_i\|}\,\hat{\mathbf{I}}\Big) \\ \Big(\sqrt{\|\mathbf{f}_i\|}\,\hat{\mathbf{I}}\Big)^\mathsf{T}\,{}^F\hat{\mathbf{H}}_I^\mathsf{T}\,\mathbf{L}_3\Big(\mathbf{p}_i/\hat{\ell},\hat{\mathbf{e}}\Big)\,{}^F\hat{\mathbf{H}}_I\,\Big(\sqrt{\|\mathbf{f}_i\|}\,\hat{\mathbf{I}}\Big) \end{bmatrix}$$

$$= \|\mathbf{f}_i\| \begin{bmatrix} \hat{\mathbf{I}}^\mathsf{T}\,{}^F\hat{\mathbf{H}}_I^\mathsf{T}\,\mathbf{L}_1\Big(\mathbf{p}_i/\hat{\ell},\hat{\mathbf{e}}\Big)\,{}^F\hat{\mathbf{H}}_I\,\hat{\mathbf{I}} \\ \hat{\mathbf{I}}^\mathsf{T}\,{}^F\hat{\mathbf{H}}_I^\mathsf{T}\,\mathbf{L}_2\Big(\mathbf{p}_i/\hat{\ell},\hat{\mathbf{e}}\Big)\,{}^F\hat{\mathbf{H}}_I\,\hat{\mathbf{I}} \\ \hat{\mathbf{I}}^\mathsf{T}\,{}^F\hat{\mathbf{H}}_I^\mathsf{T}\,\mathbf{L}_3\Big(\mathbf{p}_i/\hat{\ell},\hat{\mathbf{e}}\Big)\,{}^F\hat{\mathbf{H}}_I\,\hat{\mathbf{I}} \end{bmatrix}$$

$$= \|\mathbf{f}_i\|\,\hat{\mathbf{f}}\Big(\mathbf{p}_i, \hat{\mathbf{I}}\Big)$$

Comparing the two gives

$$\hat{\mathbf{f}}\Big(\mathbf{p}_i, \hat{\mathbf{I}}\Big) = \hat{\mathbf{r}}$$

$\hat{\mathbf{f}}(\mathbf{p}_i, \hat{\mathbf{I}})$ is called the unit force, and the corresponding current $\hat{\mathbf{I}}$ the unit current.

Step 2

The desired force $\mathbf{f}_d$ is split into magnitude and direction:

$$\mathbf{f}_d = \|\mathbf{f}_d\|\,\hat{\mathbf{r}}(\varphi,\theta)$$

$$\mathbf{I}_{\text{opt}} = \arg\min_{\mathbf{I}} \ \|\mathbf{I}\|^2 \qquad \text{s.t.} \quad \mathbf{f}(\mathbf{p}_i,\mathbf{I}) = \mathbf{f}_d(\mathbf{p}_i,\mathbf{I}_{\text{opt}})$$

Replacing $\mathbf{I}$ by $\mathbf{I} = \sqrt{\|\mathbf{f}_i\|}\,\hat{\mathbf{I}}$:

$$\|\mathbf{I}\|^2 = \|\mathbf{f}_i\|\,\|\hat{\mathbf{I}}\|^2, \qquad \mathbf{f}(\mathbf{p}_i,\mathbf{I}) = \|\mathbf{f}_i\|\,\hat{\mathbf{f}}\Big(\mathbf{p}_i,\hat{\mathbf{I}}\Big) = \mathbf{f}_d$$

The constraint gives $\|\mathbf{f}_i\| = \|\mathbf{f}_d\|$ and $\hat{\mathbf{f}}(\mathbf{p}_i,\hat{\mathbf{I}}) = \mathbf{f}_d/\|\mathbf{f}_d\| = \hat{\mathbf{r}}$, where $\|\mathbf{f}_d\|$ is a constant.
The problem becomes the minimum-norm unit-current problem:

$$\hat{\mathbf{I}}_{\text{opt}} = \arg\min_{\hat{\mathbf{I}}} \ \|\hat{\mathbf{I}}\|^2 \qquad \text{s.t.} \quad \hat{\mathbf{f}}\Big(\mathbf{p}_i,\hat{\mathbf{I}}\Big) = \hat{\mathbf{r}}$$

Solution of the original problem:

$$\mathbf{I}_{\text{opt}} = \sqrt{\|\mathbf{f}_d\|}\,\hat{\mathbf{I}}_{\text{opt}}$$

Step 3

For a fixed position $\mathbf{p}_i$, $\hat{\mathbf{I}}_{\text{opt}}(\mathbf{p}_i,\varphi,\theta)$ is solved for an arbitrary direction $\hat{\mathbf{r}}$ with Lagrange multipliers. Let

$$\mathbf{Q}_k = {}^{F}\hat{\mathbf{H}}_I^{\mathsf{T}}\,\mathbf{L}_k\Big(\mathbf{p}_i/\hat{\ell},\hat{\mathbf{e}}\Big)\,{}^{F}\hat{\mathbf{H}}_I, \qquad k = 1,2,3$$

The k-th component of the constraint $\hat{\mathbf{f}}(\mathbf{p}_i,\hat{\mathbf{I}}) = \hat{\mathbf{r}}$ is $\hat{\mathbf{I}}^{\mathsf{T}}\mathbf{Q}_k\hat{\mathbf{I}} = r_k$; with the Lagrange multipliers $\boldsymbol{\lambda} = \begin{bmatrix}\lambda_1 & \lambda_2 & \lambda_3\end{bmatrix}^{\mathsf{T}}$:

$$\mathcal{L}\Big(\hat{\mathbf{I}},\boldsymbol{\lambda}\Big) = \|\hat{\mathbf{I}}\|^2 - \sum_{k=1}^{3}\lambda_k\left(\hat{\mathbf{I}}^{\mathsf{T}}\mathbf{Q}_k\hat{\mathbf{I}} - r_k\right)$$

Step 4

With $\hat{\mathbf{I}}_{\text{opt}}$ obtained from the Lagrange multipliers, the normalized current is

$$\bar{\mathbf{I}} = \frac{\hat{\mathbf{I}}_{\text{opt}}}{\left\|\hat{\mathbf{I}}_{\text{opt}}\right\|}, \qquad \|\bar{\mathbf{I}}\| = 1$$

Substituting a current of magnitude $\|\mathbf{I}\|$ (A), $\mathbf{I} = \|\mathbf{I}\|\,\bar{\mathbf{I}}$, into the force model:

$$\mathbf{f}(\mathbf{p}_i,\mathbf{I}(\varphi,\theta)) = \|\mathbf{I}\|^2\begin{bmatrix}\bar{\mathbf{I}}^{\mathsf{T}}\mathbf{Q}_1\bar{\mathbf{I}}\\ \bar{\mathbf{I}}^{\mathsf{T}}\mathbf{Q}_2\bar{\mathbf{I}}\\ \bar{\mathbf{I}}^{\mathsf{T}}\mathbf{Q}_3\bar{\mathbf{I}}\end{bmatrix}$$

$$= \frac{\|\mathbf{I}\|^2}{\left\|\hat{\mathbf{I}}_{\text{opt}}\right\|^2}\begin{bmatrix}\hat{\mathbf{I}}_{\text{opt}}^{\mathsf{T}}\mathbf{Q}_1\hat{\mathbf{I}}_{\text{opt}}\\ \hat{\mathbf{I}}_{\text{opt}}^{\mathsf{T}}\mathbf{Q}_2\hat{\mathbf{I}}_{\text{opt}}\\ \hat{\mathbf{I}}_{\text{opt}}^{\mathsf{T}}\mathbf{Q}_3\hat{\mathbf{I}}_{\text{opt}}\end{bmatrix} = \frac{\|\mathbf{I}\|^2}{\left\|\hat{\mathbf{I}}_{\text{opt}}\right\|^2}\,\hat{\mathbf{f}}\Big(\mathbf{p}_i,\hat{\mathbf{I}}_{\text{opt}}\Big) = \frac{\|\mathbf{I}\|^2}{\left\|\hat{\mathbf{I}}_{\text{opt}}\right\|^2}\,\hat{\mathbf{r}}(\varphi,\theta)$$

Define

$$\rho(\mathbf{p}_i,\hat{\mathbf{r}}) = \frac{1}{\left\|\hat{\mathbf{I}}_{\text{opt}}\right\|^2} \qquad (\text{pN/A}^2)$$

Substituting back into the force:

$$\mathbf{f}(\mathbf{p}_i, \mathbf{I}(\varphi, \theta)) = \|\mathbf{I}\|^2 \, \rho(\mathbf{p}_i, \hat{\mathbf{r}}) \, \hat{\mathbf{r}}(\varphi, \theta)$$

$\rho(\mathbf{p}_i, \hat{\mathbf{r}})$ is the maximum force gain (pN/A^2) attainable at position $\mathbf{p}_i$ along the direction $\hat{\mathbf{r}}$.

Step 5

$$\mathcal{C}_f(\mathbf{p}_i) = \left(\frac{1}{4\pi} \int_{-\pi/2}^{\pi/2} \int_0^{2\pi} \rho^3(\mathbf{p}_i, \hat{\mathbf{r}}(\varphi, \theta)) \cos\varphi \, d\theta \, d\varphi \right)^{1/3} \qquad (\text{pN/A}^2)$$

$$\kappa_f(\mathbf{p}_i) = \frac{\min\limits_{\hat{\mathbf{r}}} \ \rho(\mathbf{p}_i, \hat{\mathbf{r}})}{\max\limits_{\hat{\mathbf{r}}} \ \rho(\mathbf{p}_i, \hat{\mathbf{r}})}$$